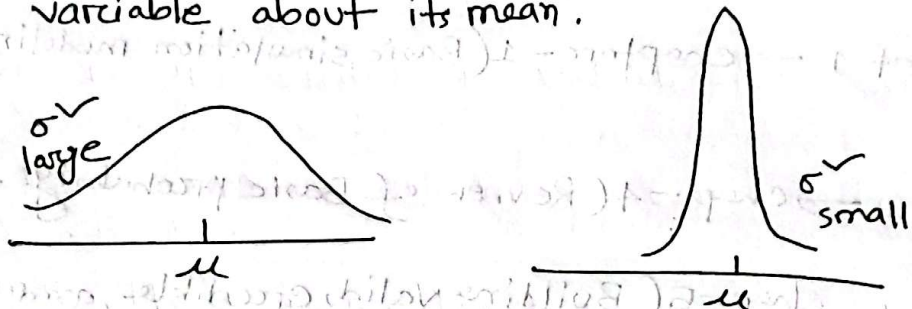


Chapter 4 and 5 → 1 set

* How to compute variance, covariance, and correlation for simulation study.

Variance

The variance is a measure of the dispersion of a random variable about its mean.



$$\sigma^v = E(x_i^2) - \mu_i^2$$

Co-Variance

The co-variance is the measure of dependence between two random variables. If the two random variables are X_i and X_j ,

Then the co-variance,

$$C_{ij} = E(X_i X_j) - \mu_i \mu_j$$

Correlation

Correlation is a statistical measure that describes the relationship between two random variables.

The correlation ρ_{ij} is defined by,

$$\rho_{ij} = \frac{c_{ij}}{\sqrt{\sigma_i^2 \sigma_j^2}}$$

* Why stochastic process is used to simulate the output data

A stochastic process is used to simulate situations where outcomes are uncertain or random, it helps predict possible ~~outcome~~ results when we can't know for sure

what will happen, but we can estimate based on

probabilities.

→ **Example** यदि कर्म शक व आये गेले example यह

Examples: Imagine you are trying to guess how many customers visit a shop each day. Some days there are 20, other days 50, and it changes randomly. A stochastic process helps simulate these changes, giving you a range of possible outcomes instead of just one fixed number.

* Illustrate correlation function of the process of delays in queue $D_1, D_2, \dots$ for the $M/M/1$ queue.

The $M/M/1$ model is a queueing process in which customers arrive at one server and wait in a queue until the server is available customers are serviced in the order in which they arrived.

The server services at most one customer at a time. There is no limit to the customer number of customer who can wait in the queue.

↓ Consider the output process D_1, D_2 for a covariance stationary $M/M/1$ queue with $\rho = \lambda/\omega < 1$.

Here λ = the arrival rate

ω = service rate

From the results in Delay (1968), one can compute ρ_j ,

which we plot in the figure for $\rho = 0.5$ and 0.9

The correlation P_j are positive and monotonically decrease to zero as j increases. In particular, $P_1 = 0.99$ for $p = 0.9$ and $P_1 = 0.78$ for $p = 0.5$. The convergence of P_j to zero is considerably slower for $p = 0.9$

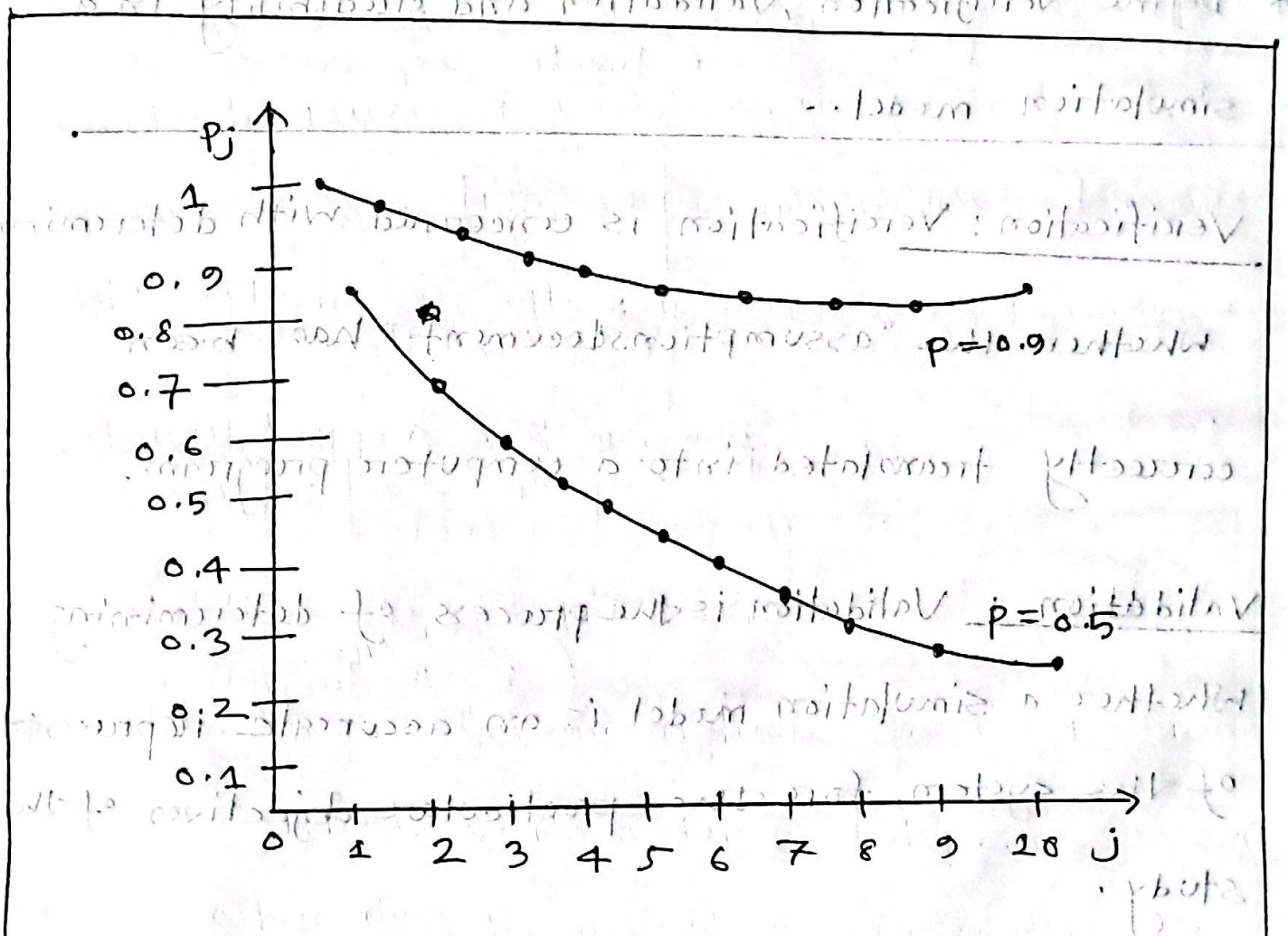


Figure: correlation function P_j of the process D_1, D_2

... for the M/M/1 queue, as detailed in

* What is random variable?

A random variable is a function (or rule) that assigns a ~~real~~ real number (any number greater than $-\infty$ and less than ∞) to each point in the sample space S .

* Define verification, validation and credibility in a simulation model.

Verification: Verification is concerned with determining whether the "assumptions document" has been correctly translated into a computer program.

Validation: Validation is the process of determining whether a simulation model is an accurate representation of the system, for the particular objectives of the study.

Credibility: Credibility refers to how trustworthy and believable the simulation results are.

* Mention the issues in an accreditation decision. (1)

1. Verification and validation that have been done.
2. Credibility of the model.
3. Simulation model development and use history.
4. Quality of the data that are available.
5. Quality of the documentation.
6. know problems or limitation with simulation model.

* Show the timing relationship between validation, verification and establishing credibility.

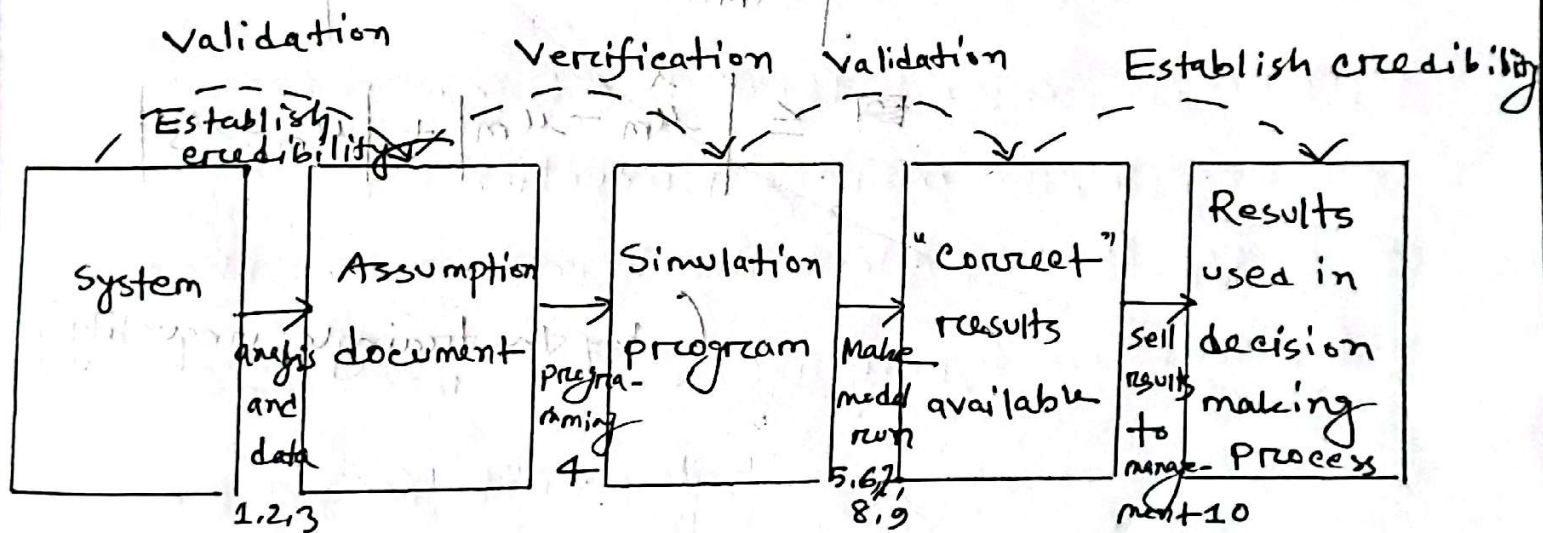


Figure: Timing relationship of validation, verification and establishing credibility.

* Derive the mathematical formulation in a good estimate of the mean of the system using both validation and output analysis (2).

suppose, we want to estimate the mean μ_s of some system. We construct a simulation model whose corresponding mean is μ_m . We make a simulation run and obtain an estimate $\hat{\mu}_m$ of μ_m .

$$\begin{aligned} \text{Error in } \hat{\mu}_m &= |\hat{\mu}_m - \mu_s| \\ &= |\hat{\mu}_m - \mu_m + \mu_m - \mu_s| \\ &\leq |\hat{\mu}_m - \mu_m| + |\mu_m - \mu_s| \end{aligned}$$

(by the triangle inequality)

→ chap-04

* Explain hypothesis testing including null and alternative hypothesis for mean. (2)

Null hypothesis (H_0):

There is no effect in the population. It is the starting assumption in statistics. It says there is no relationship between groups. For example, A company claim its average production is 50 units per day, then;

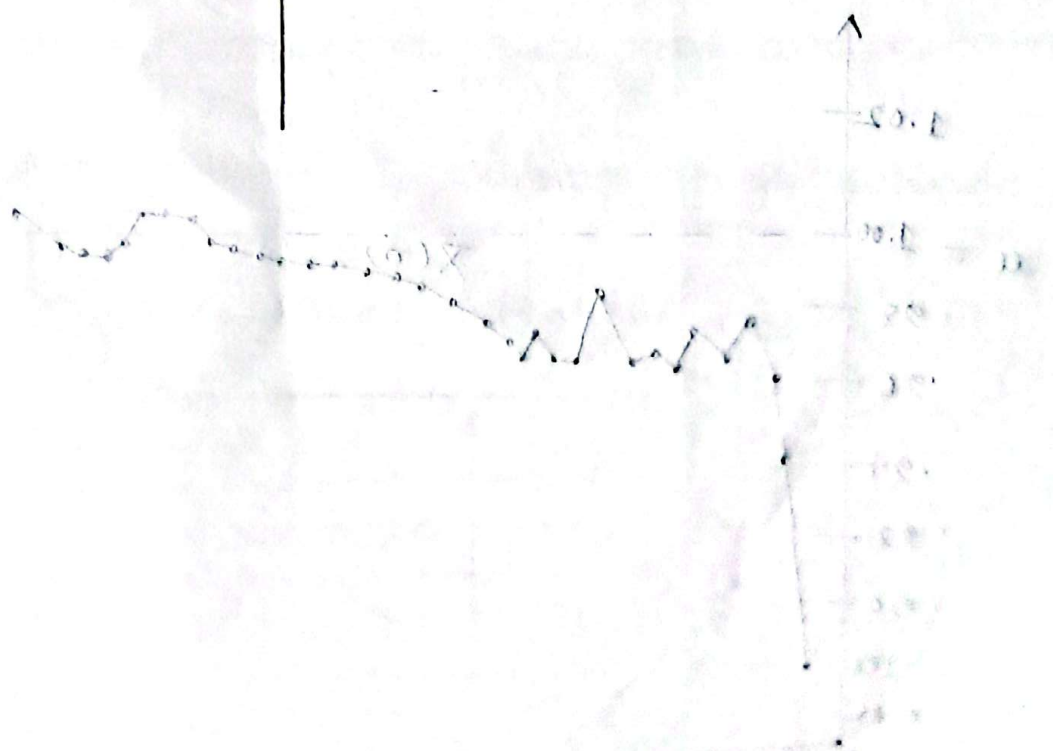
Null hypothesis (H_0): The mean number of daily visits (μ) = 50.

Alternative hypothesis (H_1): There is an effect in the population. It suggests there is a difference between group. For example, the company production is not equal to 50 units per day then alternative hypothesis would be;

Alternative hypothesis (H_1): The mean number of day visits (μ) $\neq$ 50

* What is the difference between type I error and type II error. (2)

Type I	Type II
1) If one rejects H_0 when in fact, it is true, this is called Type I error.	2) If one fails to reject H_0 when it is false, this is called a Type II error.
2) False positive	3) False negative
3) Symbol: α (alpha)	3) Symbol: β (beta)



* Explain and illustrate the strong law of large numbers.

Let $X_1, X_2, \dots$ are IID random variables with

$\mu = 1$ and $\sigma^2 = 0.01$. In this below figure the

values of $\bar{X}(n)$ for various n that resulted from

sampling from this distribution. Note that

$\bar{X}(n)$ differed from μ by $\leq$ less than 1 percent

for $n \geq 28$.



$\bar{X}(n)$ for various values of n when the X_i are normal random variable

* Explain the inspection approach of statistical procedure for comparing real-world observation and simulation output analysis. (3)

For a real-world system of interest is the M/M/1 queue with $p = 0.6$ and that the corresponding simulation model is the M/M/1 queue with $p = 0.5$, in both cases arrival rate is 1 and the output process of interest is

$D_1, D_2, \dots$

Let, $X = \frac{\sum_{i=1}^{200} D_i}{200} \rightarrow$ for the system

$Y = \frac{\sum_{i=1}^{200} D_i}{200} \rightarrow$ for the model.

We shall attempt to determine how good a representation

the model is for the system for comparing an estimate

for $\mu_Y = E(Y) = 0.49$. with an estimate $\mu_X = E(X) = 0.87$

Experiment	$\hat{\mu}_X$	$\hat{\mu}_Y$	$\hat{\mu}_X - \hat{\mu}_Y$
1	0.90	0.70	0.20
2	0.70	0.71	-0.01
3	1.08	0.35	0.73

Table is the results for three experiment with the inspection approach.